CHANDU PRASAD BHAIRAPU



AI Engineer

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ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in AI-driven applications for process environments, from machine learning on production data to generative AI that supports data-driven decision-making. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector, backed by chemical-industry experience at Covestro.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present Muldestausee, Germany

- Owned the knowledge-extraction pipeline end-to-end as sole developer, from data ingestion and chunking to prompt engineering and indexing, reporting directly to the CTO and working with SAP domain experts across the German energy sector to translate their domain requirements into extraction logic.
- Built the data pipelines feeding a Neo4j knowledge graph, with data integrity checked against an SAP-system-derived ground truth; the multi-agent extraction in LangGraph raised precision/recall from 33%/45% to 93%/90%.
- Engineered a deterministic, code-based extraction pipeline for a second, high-volume source stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing accuracy toward 100%.
- Developed Expert Interface, a curation web app (React frontend, Python microservices backend) on Azure where domain experts validate and approve every change before it reaches the knowledge graph.
- Built a Model Context Protocol (MCP) service that exposes the knowledge graph for millisecond-latency retrieval to AI tools.
- **Tech stack: Python, LangGraph, LangChain, LiteLLM, Neo4j, MongoDB, Postgres, RAG, GraphRAG, MCP, FastAPI, Docker, Kubernetes, Azure, React**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 Leverkusen, Germany

- Consulted business stakeholders on their decision-making workflows, advised where agentic AI could support them and translated their requirements into the solution approach.
- Designed and deployed an agentic AI application with LangGraph that routes requests across vector database queries, general inference and summarization; Python/FastAPI backend and React frontend.
- Containerized the stack with Docker and deployed it to AWS with CDK (IaC); versioned in GitLab with CI/CD pipelines.
- **Tech stack: Python, LangGraph, LangChain, FastAPI, Docker, AWS (CDK, EC2, Lambda), GitLab CI/CD, React**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 Leverkusen, Germany

- Automated evaluation of the PrismaChat RAG system with RAGAS and PytrecEval against chemist-provided ground truth, on AWS infrastructure (Lambda, SQS, DynamoDB) built with CDK.
- Benchmarked multilingual embedding models on over 1M chunks of internal chemical data and designed RAG pipelines (chunking, vector databases, indexing, fine-tuning models for retrieval).
- **Tech stack: Python, RAGAS, PytrecEval, LangChain, Hugging Face Transformers, PyTorch, AWS (Lambda, SQS, DynamoDB, CDK), Azure AI Search**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost and LSTM models to predict prepolymer synthesis outcomes (viscosity, reaction progression) from time-series data of chemical manufacturing production processes; benchmarked against empirical curve-fitting, the models improved early identification of batches likely to deviate from spec.
- Engineered lag features and rolling statistical windows from sensor time series for the tree models and fed raw sequences into the LSTM.
- Delivered a Streamlit dashboard for process improvement, where chemists simulate parameter changes and see predictions before running physical batches.
- **Tech stack: Python, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, NumPy, SciPy, Matplotlib, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XGBoost regressor, FB Prophet and LSTM models for EV load time-series forecasting; the LSTM reached 96% test accuracy with strong performance on unseen data.

PROJECTS

Project: Plant Maintenance Agent | Python | Scikit-Learn | RAG | FastAPI

- Predictive maintenance proof of concept on simulated plant sensor data (pump pressure, temperature, vibration) in an OPC UA-style tag model: anomaly detection model trained and evaluated on injected fault scenarios.
- Generative AI assistant for operators: RAG over equipment manuals, and an agent links a detected anomaly to the relevant manual section and proposes a maintenance action for the operator to confirm; FastAPI services in Docker with an evaluation set in CI.

TECHNICAL SKILLSET

- **Machine Learning & Advanced Analytics:** Scikit-Learn, XGBoost, PyTorch, TensorFlow/Keras, time-series forecasting (ARIMA, Prophet, LSTM), feature engineering, Pandas, NumPy, SciPy, Matplotlib, Streamlit
- **Generative AI:** LangGraph, LangChain, LiteLLM, RAG, GraphRAG, MCP, prompt engineering, RAGAS, Hugging Face Transformers
- **Data Engineering:** data pipelines, SQL, Postgres, MongoDB, Neo4j, vector databases (Qdrant, Pinecone, Azure AI Search)
- **Cloud Infrastructure & DevOps:** IaC (AWS CDK), Azure, AWS (EC2, Lambda, SQS, DynamoDB, S3, CDK), Docker, Kubernetes, Git, GitLab CI/CD, FastAPI, React (beginner)
- **Ways of working:** stakeholder management and consulting on custom AI solutions, Agile, AI-assisted development with Claude Code

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering — August 2017 – July 2021
Minor in Computer Science & Engineering
G Pulla Reddy Engineering College, Kurnool, India

PUBLICATIONS & CERTIFICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles
- Udemy - Machine Learning A-Z : AI, Python + ChatGPT
- Udemy - LLMs Mastery : Transformers & Generative AI

LANGUAGES

English (Business Fluent) | German (B1) | Hindi (Fluent) | Telugu (Native)